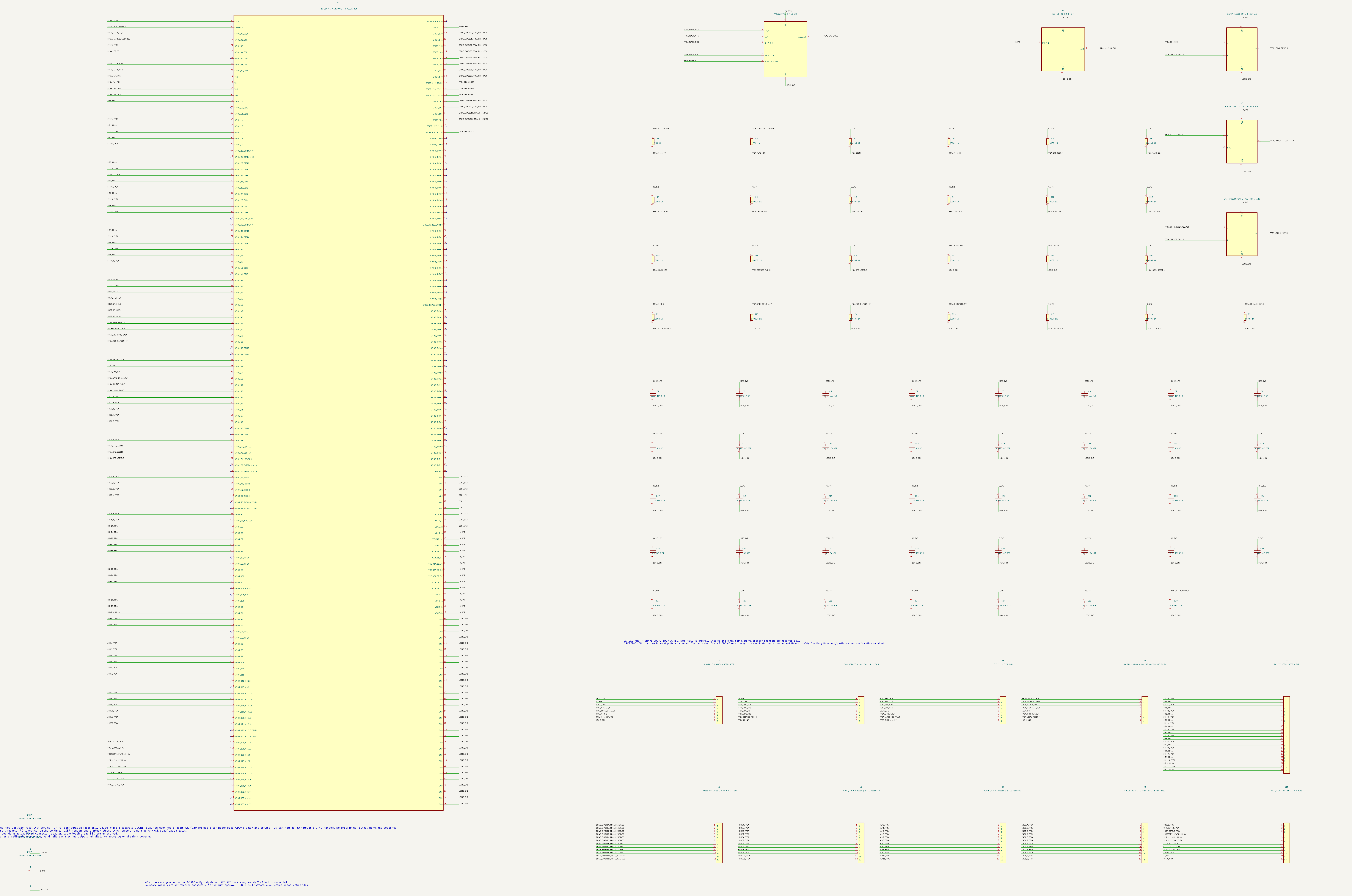
T20F256 FPGA CORE / MINIMUM 8 MOTOR CHANNELS, 12—CHANNEL PIN CAPACITY CANDIDATE

Exact 256 balls: Efinix T20—PINOUT worksheet T20F256. Pin allocation is a board candidate, NOT vendor fit or routed timing.
All VCCIO=3.3V. All VCC/VCCA=1.2V. No 5V—tolerant FPGA connection. No native LVDS: REF_RES intentionally NC.

One 100nF per supply ball; bulk per physical supply group. Decoupling is a layout—study allowance, not a PDN proof.
Active SPI x1 straps111; CSI/TEST high. Flash bus reserved from user RTL. CCK damping33R is provisional.
CDONE is open—drain with10k pullup; status does not grant motion permission. Configuration clock is internal, not Y1.


U1
T20F256I4 / CANDIDATE PIN ALLOCATION
U2
W25Q32JVSSIQ / x1 SPI
Y1
ASE—50.000MHZ—L—C—T
U3
SN74LVC1G08DCKR / RESET AND
U4
74LVC1G17GW / CDONE DELAY SCHMITT
U5
SN74LVC1G08DCKR / USER RESET AND
J1
POWER / QUALIFIED SEQUENCER
J2
JTAG SERVICE / NO POWER INJECTION
J3
HOST SPI / 3V3 ONLY
J4
HW PERMISSION / NO ESP MOTION AUTHORITY
J5
TWELVE MOTOR STEP / DIR
J6
ENABLE RESERVES / CIRCUITS ABSENT
J7
HOME / 0—5 PRESENT; 6—11 RESERVED
J8
ALARM / 0—5 PRESENT; 6—11 RESERVED
J9
ENCODERS / 0—1 PRESENT; 2—3 RESERVED
J10
AUX / EXISTING ISOLATED INPUTS
#FLG01
SUPPLIED BY UPSTREAM
#FLG03
SUPPLIED BY UPSTREAM
CORE_1V2
IO_3V3
LOGIC_GND
FPGA_CLK_SOURCE
FPGA_CRESET_N
FPGA_SERVICE_RUN_N
FPGA_LOCAL_RESET_N
FPGA_USER_RESET_DELAYED
FPGA_USER_RESET_N

J1—J10 ARE INTERNAL LOGIC BOUNDARIES, NOT FIELD TERMINALS. Enables and extra home/alarm/encoder channels are reserves only.
CRESET47k/1k plus two internal pullups screened. The separate 10k/1uF CDONE reset delay is a candidate, not a guaranteed time or safety function; threshold/partial—power confirmation required.

RESET: U3 combines qualified upstream reset with service RUN for configuration reset only. U4/U5 make a separate CDONE—qualified user—logic reset; R22/C39 provide a candidate post—CDONE delay and service RUN can hold it low through a JTAG handoff. No programmer output fights the sequencer.
User—logic reset release threshold, RC tolerance, discharge time, tUSER handoff and startup/release synchronizers remain bench/HDL qualification gates.
J2 is a logical service boundary: actual keyed connector, adapter, cable loading and ESD are unresolved.
JTAG programming requires a deliberate reset cycle, valid rails and machine outputs inhibited. No hot—plug or phantom powering.

NC crosses are genuine unused GPIO/config outputs and REF_RES only; every supply/GND ball is connected.
Boundary symbols are not released connectors. No footprint approval, PCB, DRC, bitstream, qualification or fabrication files.

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